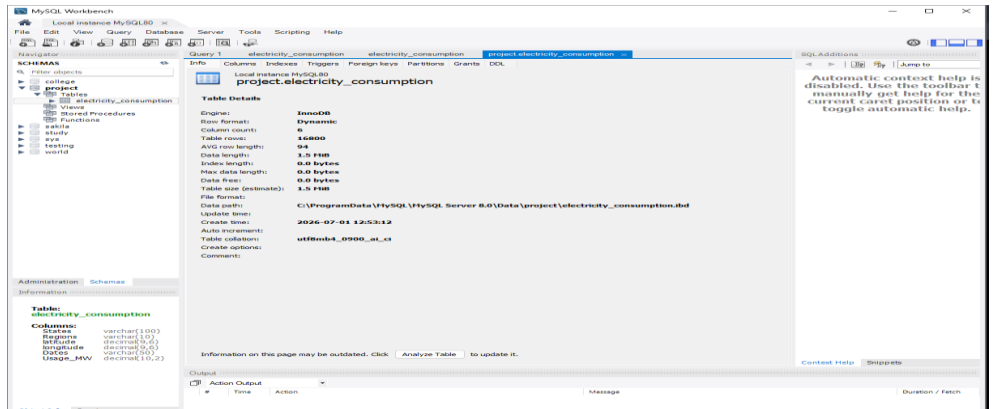


PROJECT PERFORMANCE TEST

Date	7 July 2026
Team ID	SWTID-2026-2886
Project Name	Plugging into the Future: Electricity Consumption Patterns
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.N o.	Parameter	Screenshot / Values
1.	Data Rendered	File Name: Consumption India.csv File Type: CSV (Text File Connection) Total Records: 16,800+ time-series entries Total Columns: 6 <i>Status:</i> Dataset successfully imported into Tableau Public using the native Text File connection interface. 
2.	Data Preprocessing	Missing Values: 0 (No missing values found) Duplicate Records: 0 (No duplicate records found) Data Types Verified: Date (for timestamps), String (States and Regions), Geolocation (Latitude & Longitude), and Continuous Number/Float (Usage MW).

State	Region	Latitude	Longitude	Date	Usage
Punjab	NR	31.51977028	75.58000281	2/7/2019	119.9
Haryana	NR	28.40000633	77.01999101	2/7/2019	130.3
Rajasthan	NR	26.46999921	74.61998124	2/7/2019	234.1
Delhi	NR	28.46999929	77.20000403	2/7/2019	85.4
UP	NR	27.59998089	78.05000045	2/7/2019	313.9
Uttarakhand	NR	30.12000095	78.05000045	2/7/2019	40.7
MP	NR	31.10002545	77.56659704	2/7/2019	30
J&K	NR	33.45	76.24	2/7/2019	52.5
Chhattisgarh	NR	30.71998087	76.70000045	2/7/2019	5
Gujarat	WR	22.09420355	82.11998734	2/7/2019	78.7
MP	WR	22.2587	74.1354	2/7/2019	319.5
MP	WR	23.30001005	76.10001049	2/7/2019	251
Maharashtra	WR	19.25021195	73.58017483	2/7/2019	426.6
Goa	WR	15.4912957	73.82000045	2/7/2019	12.8
DNH	WR	26.26657819	73.0166178	2/7/2019	18.6
Andhra Pradesh	SR	14.7506291	78.57002259	2/7/2019	164.6
Telangana	SR	18.1124	78.0251	2/7/2019	206.2
Karnataka	SR	12.57038129	76.51999711	2/7/2019	206.1
Kerala	SR	8.900372741	76.59999283	2/7/2019	72.7
Tamil Nadu	SR	12.5000176	79.1000487	2/7/2019	268.1
Pondy	SR	11.91999371	79.80000037	2/7/2019	6.3
Bihar	ER	25.70344405	87.1799127	2/7/2019	82.1
Jharkhand	ER	23.80001849	86.4199572	2/7/2019	24.8
Odisha	ER	19.8242971	85.90001746	2/7/2019	70.2
West Bengal	ER	22.58001044	88.10994605	2/7/2019	108.2
Sikkim	ER	27.1313103	88.4166475	2/7/2019	2
Arunachal Pradesh	NER	27.10000878	93.5566071	2/7/2019	2.1
Assam	NER	26.74998089	94.21666784	2/7/2019	21.7
Manipur	NER	24.79997072	93.95001705	2/7/2019	2.7
Meghalaya	NER	25.17063117	91.8000142	2/7/2019	6.1
Mizoram	NER	23.71008899	92.72001461	2/7/2019	1.9
Nagaland	NER	25.4899979	94.11657019	2/7/2019	2.2
Tripura	NER	23.82454248	91.27999934	2/7/2019	3.4
Punjab	NR	31.51977028	75.58000281	3/7/2019	121.9
Haryana	NR	28.40000633	77.01999101	3/7/2019	135.1
Rajasthan	NR	26.46999921	74.61998124	3/7/2019	240.2
Delhi	NR	28.46999929	77.20000403	3/7/2019	85.4

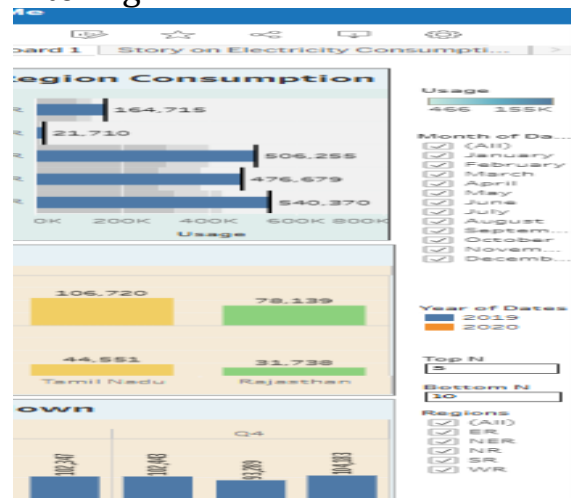
Splitting/Merging: Not Performed

Relationships: Not Required (Unified single time-series CSV dataset).

3. Utilization of Filters

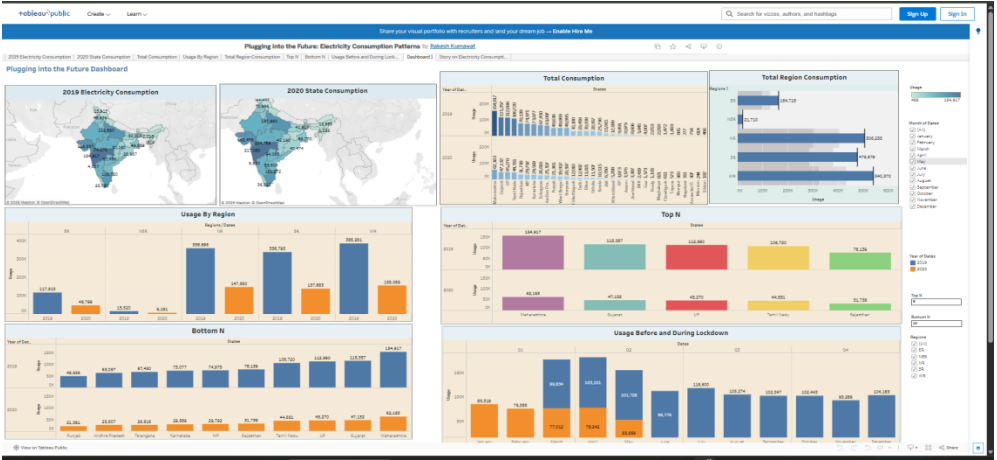
Filters used across different worksheets: Year of Dates (2019 vs 2020), Month of Dates (January to December), Regions (ER, NER, NR, SR, WR), Top N, and Bottom N parameters.

Aggregation functions: SUM(Usage) was primarily used for total load mappings, and AVG(Usage) was applied to monitor ongoing baseline shifts. Worksheets utilize coordinated dynamic action filters to allow synchronized dashboard cross-filtering.



4. Calculation fields Used

Calculated Field Name: Top N Filter / Bottom N Filter

		Logic / Syntax: Mapped via custom integer parameters linked with rank/index functions (INDEX()) to dynamically filter states based on the selection. Purpose: Solves Tableau's order of operations conflict, allowing the dashboard to dynamically rank the top or bottom states within specific geographical region filters without breaking.
5.	Dashboard design	Number of Dashboards: 1 (Plugging into the Future Dashboard) Number of Visualizations/Graphs: 8 dynamic panels (2 Interactive Maps, 5 Distinct Bar Charts, and 1 Combined Time-Series Line/Area Grid). Status: Fully interactive interface configured with cross-sheet action filters, color-coded legends, and sliding parameters for seamless business analysis. 
6	Story Design	Number of Stories: 1 (Story on Electricity Consumption) Number of Visualizations/Graphs: 8 panels mapped across sequential narrative tabs. Status: Tableau Story created to walk users step-by-step through the 2019 baseline load, the 2020 lockdown emergency drop, and individual state stress points. -

